DSB perception: BEVFusion (detection + BEV segmentation) and FlashOCC (occupancy)

BEVFusion: camera Swin-T + LSS/DepthLSS, LiDAR SparseEncoder, ConvFuser; det head TransFusion, seg head 6-class BEV. FlashOCC: ResNet-50 + LSS view transform + 2D BEV encoder + channel-to-height occupancy head.



6 cameras · real val frame

BEVFusion | camera + LiDAR

Swin-T + GeneralizedLSSFPN

DepthLSS / LSS view transform

camera BEV 80×180×180

TransFusion head 3D detection



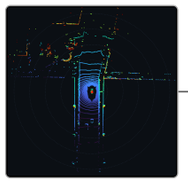
3D detection: BEV + camera

Extrinsics / aug
camera2lidar · lidar2ego
img_aug · lidar_aug

geometry

ConvFuser 80+256 → 256

SECOND + SECONDFPN 512×180×180



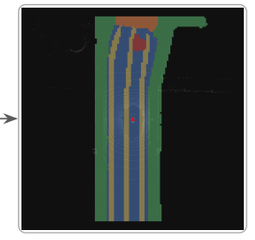
LiDAR point cloud (BEV)

Voxelize 0.075 m

SparseEncoder spconv

LiDAR BEV 256×180×180

BEV seg head 6-class map



BEV segmentation: 6 classes



6 cameras + 7-tuple geometry

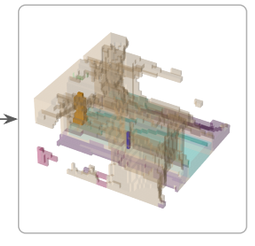
ResNet-50 + CustomFPN

LSSViewTransformer 64×200×200

CustomResNet + FPN_LSS 256×200×200

BEVOCCHead2D 18 × 16 channels

FlashOCC | camera only



occupancy: 200×200×16

Grids: det 0.075 m voxel / 180×180 BEV; seg 0.1 m voxel / 128×128 BEV (+ 200×200 map @ 0.5 m); occ 0.4 m / 200×200×16 occupancy. BEVFusion uses LiDAR + 6 cameras; FlashOCC is camera-only.